

Ankur Aditya

☎ (413) 406-8078 • ✉ aaditya@umass.edu • 🌐 ankuraditya98.github.io
in ankuraditya13 • 📄 ankuraditya98 • 🏠 Google Scholar
🏠 Amherst, MA, USA

RESEARCH INTERESTS

Multimedia Systems, Computer Vision, Computer Networks, RF/Wireless Sensing

EDUCATION

University of Massachusetts (UMass), Amherst

Sept '22 - Present

Doctor of Philosophy (MS/PhD), Computer Science

CGPA: 3.8/4

Relevant Courseworks: Distributed and Operating Systems, Intelligent Visual Computing, Advanced Algorithms, Neural Networks, Advanced Information Assurance

Indian Institute of Technology Kanpur (IIT-K), Kanpur

Jan '21 - May '22

Master of Science (MS), Electrical Engineering

CGPA: 8.67/10

Transferred to UMass Amherst for continuation of Ph.D.

Relevant Courseworks: MIMO Wireless Communications, Detection and Estimation Theory, Error Correcting Codes, Representation and Analysis of Random Signals, Machine Learning for Wireless Communications

International Institute of Information Technology (IIIT), Bhubaneswar

Aug '16 - June '20

Bachelor of Technology (B.Tech), Electronics and Telecommunication Engineering

CGPA: 9.06/10

Relevant Courseworks: Antenna and Wave Propagation, Mobile Communications.

HONORS AND AWARDS

- Recipient of an ACM Travel Grant for The 22nd ACM Workshop on Hot Topics in Networks (HotNets'23).
- Awarded a government-based merit scholarship for two consecutive years (2016-2020) during undergraduate studies.
- Ranked 1st in the Electronics and Telecommunication department during undergraduate studies, placing in the top 5% of the graduating class.

PUBLICATIONS

- **Ankur Aditya**, D. Maji, L. Wang, B. Ramakrishna, R. Sitaraman, P. Shenoy. 2026. ReVo: A Cross-Layer Reliable Volumetric Videoconferencing System. arXiv preprint arXiv:2604.27441.
- **Ankur Aditya** et al., DeformRF: Data-driven Beamforming and Direction Finding with Deformable Antenna Arrays (**Accepted at MobiSys'26**).
- X. Chen, **Ankur Aditya**, Z. Lei, D. Ganesan. 2023. Curtain-Net: Enabling precise beamforming with a deformable antenna array on a fabric substrate. In The 21st ACM Conference on Embedded Networked Sensor Systems (SenSys '23), November 12–17, 2023, Istanbul, Turkiye.

RESEARCH EXPERIENCE

Research Assistant, Computer Science, University of Massachusetts

May '25 - April '26

Advisor: Prof. Prashant Shenoy

Amherst, MA, USA

- **Project: ReVo: A Cross-Layer Reliable Volumetric Videoconferencing System (Under Review)**
 - Engineered a WebRTC-based cross-layer volumetric video system, combining network-layer FEC with neural recovery to reduce video freezes by 95.7% under bursty packet loss.
 - Designed modality-aware Video Vision Transformers (ViViT) to reconstruct corrupted RGB and depth frames, boosting median SSIM by up to 32% across standard and neural video codecs.
 - Deployed the end-to-end pipeline across diverse AWS networks, optimizing inference to strictly meet a 33ms real-time latency budget on desktop GPUs while sustaining > 0.91 median SSIM.

Research Assistant, Computer Science, University of Massachusetts

Advisor: Prof. Deepak Ganesan

Jan '24 - Aug '24

Amherst, MA, USA

• **Project: DeformRF: Data-driven Beamforming and Direction Finding with Deformable Antenna Arrays**

- Developed a novel predictive model using a Vision Transformer (ViT) to estimate the electromagnetic (EM) characteristics of flexible antennas.
- Demonstrated the successful application of computer vision methods to a real-world physics problem, achieving comparable accuracy to EM simulators (e.g., HFSS) but orders of magnitude faster.
- Built a comprehensive simulation library (D2EM) mapping physical deformations to EM properties, enabling a 2x improvement in physical beamforming and over 20x in large-scale simulations.

Research Assistant, Computer Science, University of Massachusetts

Advisor: Prof. Deepak Ganesan

Jan '23 - May '23

Amherst, MA, USA

• **Project: Curtain-Net: Enabling precise beamforming with a deformable antenna array on a fabric substrate**

- Explored solutions for signal degradation caused by physical deformations (curvature, bending) in flexible antenna arrays mounted on fabric.
- Developed CurtainNet, an end to end system delivering over 12dB of beamforming gain and extending wireless coverage radius by up to 20m by correcting for these deformations.

Research Intern, Electrical Engineering, Indian Institute of Technology

Advisor: Prof. K. V. Srivastava

Dec '19 - June '20

Kanpur, UP, India

• **Project: Design of Microwave Absorbers with Transmission Band using Frequency Selective Surfaces**

- Designed and simulated a novel, ultra-thin (5.57mm) metamaterial absorber with an integrated transmission band, achieving a peak insertion loss of only 0.167dB.
- Implemented an Absorption-Transmission-Absorption (A-T-A) rasorber structure, realizing two broad-band absorption bands (5.27 to 11.63 GHz, 15.35 to 17.11 GHz) and a narrow, highly selective transmission band (11.63 - 15.35 GHz).
- Verified performance and robustness through full-wave High Frequency Structure Simulator (HFSS) simulations.

TEACHING EXPERIENCE

Teaching Assistant, Computer Science, University of Massachusetts

Sep '22 - Present

Amherst, MA, USA

- Reasoning Under Uncertainty (Fall '22, Fall '23, Spring '25 and Fall '25)
- Introduction to Computation (Fall '24)
- Scalable Web Systems (Summer '23)
- Introduction To Data Science (Summer '23)

ACADEMIC SERVICES

- **Artifact Evaluation Committee**, ACM CoNext 2026
- **Shadow Reviewer**, ACM MobiCom 2024
- **Volunteer**, Graduate Admission Program Committee, Indian Institute of Technology Kanpur, India

SKILLS

- **Programming Languages:** ◇ Python ◇ MATLAB
- **Frameworks & Libraries:** ◇ PyTorch ◇ aiortc ◇ Open3D ◇ scikit-learn ◇ TorchCodec
- **Antenna Simulation Tools:** ◇ Ansys HFSS ◇ CST Microwave Studio
- **Other Tools:** ◇ Docker ◇ Git ◇ GNU Radio ◇ Blender ◇ PyRealSense ◇ MeshLab ◇ LaTeX